Test Report

Real Estate Project Feasibility and Financial Appraisal Tool. 30 October 2026, updated 5 November 2026 (LLM memo path tested with a real key). Everything below was executed on the development machine (Linux, Python 3.12, Streamlit 1.66, LibreOffice Calc, Google Chrome). Commands to reproduce are given for each layer.

1. Summary

Layer	What	Count	Result
Unit tests	NPV, IRR, payback, weights, base-case identities	10	pass
Property tests	Accounting identities on 200 random input sets; NPV and IRR against `numpy_financial` on 60; IRR-is-a-root and break-even consistency on 60; monotonicity on 40; no-debt equality on 20	361	pass
Edge-case tests	Invalid inputs rejected, 100% debt with zero sweep, all cash at booking, zero booking, single-month windows, loss-making case, zero interest, late land, sales beyond horizon, sensitivity shape, memo number check	12	pass
Browser test	27 checks against the running app through Chrome, including a live LLM memo	27	pass
LLM memo	Base-case memo generated by Claude Sonnet 5.5 and Claude Opus 5.5 through the AIML API; every number checked against the fact sheet	2 memos, 0 unmatched numbers	pass
Excel benchmark, base case	17 summary metrics and 7 monthly columns against a LibreOffice recalculation	17 + 7	all match within 5e−14
Excel benchmark, extra cases	3 input sets × 10 metrics through the same formulas	30	all match; 2 IRRs flagged as non-unique, as designed
Decimal recomputation	36 months of equity cash flow and NPV from the written assumptions	37	match within 1.5e−14
Performance	Full page recompute after an input change	1	1.7 s (was 27.7 s before fix)

Total automated checks: 383 pytest + 27 browser + 91 benchmark comparisons + 2 LLM memo number checks.

```
source .venv/bin/activate
pytest -q                              # 383 passed in about 2 minutes
streamlit run app.py &                 # then:
python tests/ui_test.py                # 27/27, screenshots in docs/screenshots/
python benchmark/compare.py            # base case vs Excel
python benchmark/compare_cases.py      # three more cases vs Excel
python benchmark/hand_check.py         # decimal recomputation
```

2. What the browser test does

`tests/ui_test.py` drives the installed Chrome with Playwright. It loads the page, reads the KPI tiles, opens every tab, and performs real user actions. Each check and the observed value:

#	Check	Observed
1	Base case equity NPV tile	₹7.45 cr
2	Base case equity IRR tile	47.2%
3	Base case payback tile	27 mo
4	Base case break-even tile	₹6,250/sq ft
5	Verdict line reads Viable	yes
6	Cash flow tab renders the monthly table	yes
7	Cash flow tab offers CSV download	yes
8	Sensitivity tab renders table and tornado chart	yes
9	Metric radio switches to Equity IRR	yes

#	Check	Observed
10	Scenario tab shows the preset table and names Downside, Severe, Upside	yes
11	Five custom stress sliders present	yes
12	Template memo generated	starts "INVESTMENT MEMO"
13	Memo quotes the NPV	₹7.45 cr present
14	Memo recommends PROCEED for base case	yes
15	Claude mode with no key shows a handled error, not a crash	"Could not generate memo: ... authentication"
16	Typing price 6,250 gives NPV about zero	₹0.00 cr
17	Typing price 6,250 gives IRR equal to the hurdle	15.0%
18	Sold-out month 5 (before launch) shows a validation error	"Sales end month must be ≥ launch month."
19	Invalid input hides the KPI tiles	yes
20	Reset button restores the base case	₹7.45 cr
21	Price 4,000 flips the verdict to Not viable	yes
22	Loss case shows No payback	yes
23	Sold-out month 40 extends the horizon to 46 months	46 months
24	Full recompute after a change under 6 s	1.7 s
25	Construction cost 3,300 lowers NPV	₹5.52 cr

Screenshots: `docs/screenshots/01_summary.png` to `08_not_viable.png`, plus `03b_sensitivity_irr.png` and `05b_memo_llm_error.png`.

3. What testing found and what changed because of it

#	Found by	Finding	Severity	Change
1	First run	IRR returned None for the base case (bisection bracket)	High	Grid scan then bisection
2	Reading output	Loan unused; interest ₹0.04 crore	High (model realism)	Debt funds land, approvals and construction
3	Edge-case test	Sensitivity table had 15 mostly empty columns	Medium (usability)	Positional step columns
4	Edge-case test	Template memo contained a number absent from the fact sheet	Low	Cushion added to fact sheet
5	Extra Excel cases	Excel IRR() returned #N/A on the downside case	Medium (benchmark)	Guess argument in formula
6	Extra Excel cases	Two valid IRRs in the heavy-debt case; tool and Excel reported different ones	High (decision risk)	Non-unique IRR detection and on-screen warning
7	Detector on base case	First detector flagged the base case because of a root at −98% per year	Medium	Root band restricted to −10% to +50% per month; NPV-sign consistency check
8	Property tests	My assertion that NPV always falls with the rate was false for loss-making and non-conventional flows	None (model correct)	Test restricted; recorded as a learning
9	Screenshot review	Tornado colours inverted for cost drivers	Medium (misleading chart)	Colour by metric direction
10	Screenshot review	Sensitivity table too wide for its column	Low	Shorter step labels
11	Browser timing	27.7 s per input change; sensitivity loops ran two break-even solves per appraisal	High (demo would stall)	Break-even skipped inside loops; 1.7 s
12	Browser test	Edit immediately after Reset lost while widgets remounted	None (test timing)	Test helper verifies and retries
13	Tooling	Snap LibreOffice cannot write to /tmp; kill pattern matched its own shell	None	Project-local folder; kill by port

#	Found by	Finding	Severity	Change
14	LLM memo	First real memo (Sonnet 5.5 via AIML API) said the fact sheet "does not say which assumptions were changed" in the scenarios, so it could not judge their likelihood	Memo less useful	Scenario definitions added to the fact sheet; memo regenerated and re-checked
15	LLM memo	Memo reported that a 3-month construction delay *raises* NPV (₹7.82 cr vs ₹7.45 cr) and flagged it as counter-intuitive	None in code: the model has no delay penalty, so later spend is discounted more. Correct given the assumptions	Noted as a known limitation in the manual; the memo's caveat was kept
16	Browser test	On a loaded machine the "Running" status widget appeared 1.2 s after Enter, after the test had already read the stale metric; 3 checks failed although the app was right	Flaky test	Test now waits for the widget to attach before waiting for it to detach

4. What was not tested

- The Anthropic-direct provider with a real Anthropic key. Only its no-key error path was tested; the live memo was produced through the AIML API gateway (same prompt, same fact sheet, Claude models). Memo text itself was read by a person once (see `docs/llm_memo_claude-sonnet-5-5.md` and `docs/llm_memo_claude-opus-5-5.md`); LLM output varies between runs and must be re-read every time.
- Windows and macOS installation. Only Linux was used.
- (Closed 5 Nov 2026.) Streamlit Community Cloud: all 27 browser checks, including the live Claude memo, passed against https://ce725-real-estate-feasibility.streamlit.app/ (recompute 3.6 s).
- Browser rendering in Firefox or Safari; only Chrome was used.
- (Closed 5 Nov 2026.) The hand calculation by a person with a calculator is now in section 6 of the validation report, re-performed by the Validation Engineer.